Programming Fundamentals II: Lab 01

Due Date: Tuesday September 10, 2024 at Midnight

Part 1. Intro to Arrays (10 Marks)

For this week's lab, we are going to work with the basics of arrays in Java.

We'll be working with data pulled from Google's AdWords API. This data represents certain terms that have been searched on Google. We have approximate data for the search volume of each of these terms were searched for each month in 2024. Search volume here represents how often a given phrase was searched on Google. For example, in January, the term "Java" was searched approximately 450,000 times on Google. For this lab, we will be doing some basic analysis of this data.

term \ month	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY
java	246000	450000	201000	201000	201000	201000	201000
programming	40500	33100	33100	33100	33100	33100	33100
university of winnipeg	1900	1600	1600	1600	1600	1300	1300

Data Source: Google Adwords API, Sep 01, 2024

Download the Lab01.java file provided and make the following changes:

a. **(1 Mark)** Declare an array called months. This array should store each of the months shown in the data above.
b. **(2 Marks)** Declare 3 arrays called javaVols, programmingVols, and uofwVols. These arrays should store their respective monthly search volumes.
c. **(2 Marks)** Create a method named calcMax. This method should accept in an array of search volumes, and return the index of the greatest value in the array. (If there's a tie for greatest, return the first value that tied.)
d. **(2 Marks)** Create a method named calcAverage. This method should accept in an array of search volumes, and return the average of those values.
e. **(1 Mark)** Display the average monthly search volume for the term "university of winnipeg"
f. **(2 Marks)** Display the *name of the month* that the term "java" received the greatest search volume for.

Add the **date and your name and student number** at the top of the java file. Please submit your Lab01.java file to Nexus before midnight on the due date. **Late submissions will be not be accepted.**